Preet Sambare, Academic Credentials

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PROFESSIONAL SUMMARY

A mechanically driven engineer with a strong interest in vehicle dynamics, mechanical design, and system-level problem-solving. With hands-on experience spanning heavy construction machinery, competitive off-road vehicle engineering, agricultural automation, and aerospace design, bringing a practical, multidisciplinary approach to engineering challenges. My technical experiences are complemented by competitive engineering achievements and leadership roles in both technical teams and extracurricular organizations.

EDUCATION

Dr. D.Y. Patil College of Engineering (Savitribai Phule Pune University) Pune, India
Bachelors of Engineering in Mechanical Engineering, Specialization in Electric Vehicles. *2022-2026*
Thesis Title: Automated weed killer. (Role: Subsystems design, Project manager)

TECHNICAL / LAB SKILLS

Data Analysis & Software: Siemens Nx & Teamcenter (14), Catia (V5R21), IPG Carmaker (V11)
Laboratory Skills: Fixturing, Prototyping, Shaping, Machining, Laser Cutting
Certifications: Aircraft Design Training.

RELEVANT EXPERIENCE

JCB India (Design Plant) Pune, India
Design Intern at Cab systems Department. *June 2025 – August 2025*

- **Designed** 'JCB DAKSH', a virtual excavator simulator integrating physical operator control interfaces, machine-operation logic, and simulated equipment responses to replicate excavator operating dynamics for realistic, repeatable operator training in Virtual reality.
- **Enhanced** Wheel Loader (WLS) cab structural design by supporting design updates focused on operator safety, ergonomic comfort, and structural integrity.
- **Analyzed** competitor cab designs, safety standards, and emerging control technologies to identify improvement opportunities and support product benchmarking.
- **Researched** electro-hydraulic control systems for excavators, evaluating efficiency, precision, and integration with existing hydraulic architectures to inform future development.

Team Predators Racing (E-BAJA SAE INDIA) Pune, India
Design & Test Engineer for Suspension system and Vehicle dynamics. *June 2023 – May 2024*

- **Suspension Design**: Engineered SLA double-wishbone front and 3-link semi-trailing rear geometry, achieving 31% anti-dive, 71% anti-squat, and progressive motion ratios of 0.68/0.62.
- **Damper Development:** Designed a coil-over damper with 3-way adjustability (high-/low-speed compression and rebound), dual-spring tuning, and progressive damping for multi-terrain adaptability.
- **CAE & Structural Validation:** Conducted ANSYS FEA on control arms and trailing arms under bump loads up to 6.33 kN; validated component deformation using strain gauges.
- **Experimental Testing:** Validated damper behavior on an MTS test rig, measured 5.8 in (147 mm) travel, and verified front/rear spring rates of 20/25 N/mm (primary) and 27/30 N/mm (secondary).

- **Manufacturing & Failure Analysis:** Designed suspension-link fabrication fixtures and replaced a bending aluminum lateral link with steel following prototype failure, improving resistance to demanding operating loads.
- **All India Rank 1** in **Suspension and Traction Test** E-Baja SAE India 2023.

Team Interceptors (SAE TIFAN) Pune, India
Product level Design Engineer & Senior Team Manager. *January 2025 – June 2025*

- **Hitch Mechanism Design:** Engineered a tractor-implement hitch compatible with Category 1 and Category 2 three-point linkage systems, accommodating differing hitch geometries for cross-tractor integration. (The only design to achieve this nationwide.)
- **Automated Transplanter Development**: Led the development of an "*Automated Vegetable Transplanter"* capable of planting 3 sapling types, integrating modular planting mechanisms with a tractor-mounted frame for field operations within the cost and other constrains given by the Rule book.

Feynman Aerospace (Online training and Project internship program) Berlin, Germany
Design Trainee for Aircraft design program. *October 2024 – December 2024*

- **Aerodynamic Modelling:** Analysed airfoil geometry, lift coefficient, drag coefficient, and lift-to-drag ratio using XFLR5 to evaluate aerodynamic efficiency across operating condition.
- **Aircraft Design & Structural Analysis:** Developed wing, fuselage, and empennage CAD geometry in CATIA and performed ANSYS-based structural analysis to evaluate stress distribution, deformation, and structural integrity under applied loading.

TEACHER ASSISTANCE AND KNOWLEDGE TRANSFER EXPERIENCES

Dr. D.Y. Patil College of Engineering, **Department of Mechanical Engineering** Pune, India
Teacher Assistant for Design of Transmission System Course

- Delivered lectures on rolling- and sliding-contact bearings, covering bearing selection, equivalent dynamic load calculations, bearing life estimation, and load-capacity evaluation using machine design principles prominently while also coordinating the academic submissions with the faculty and students keeping an account of related things for a semester.

SERVICE & LEADERSHIP EXPERIENCE

UDBHAV – The Official Anchoring Club of DYPCOE Pune, India
Founder & President. *December 2022 – June 2026*

- Led Udbhav, DYPCOE's official anchoring club, leading event hosting, coordinating stage operations, and managing institutional programmes to deliver engaging, professionally executed events with audiences being more than 4000 at times at events held both on and off campus. While conducting campus-wide anchoring workshops, mentoring students in stage presence and public speaking.

AWARDS & HONORS

- **All India Rank - '1' *E-BAJA SAE INDIA 2023*** *(10/18 awards won in 1 season)*
- **Virtual Dynamics (All Terrain Performance) – *AIR 3***
- **All India Rank 1** in **Suspension and Traction Test** E-Baja SAE India 2023.
- **Lightest Vehicle in the entire competition** (*149 kgs with nearest next being 209 kgs*)
- **Speaker of the year – 2026 at Dr. D.Y. Patil College of Engineering.**